Linear Approximation Review

Let $F(x)$ be a differentiable function. Then the linear approximation to $F(x)$ at $x = a$ is

$$F(x) \approx _____ \text{ near } x = a.$$

Using this to estimate $F(a + \Delta x)$, we get

$$F(a + \Delta x) \approx _____ = _____.$$

So, How Does One Compute Those Areas?

Suppose $f(x)$ is continuous. We want to try to evaluate $\int_a^b f(x)\,dx$. Suppose also that $F(x)$ is an antiderivative of $f(x)$, that is, $F'(x) = f(x)$:

If we know the values of $F(a)$ and $F'(a) = f(a)$, we can apply linear approximation to approximate $F(a + \Delta x)$:

- $F(a + \Delta x) \approx _____ = _____$

- Use $F'(a + \Delta x) = f(a + \Delta x)$ to approximate $F(a + 2\Delta x)$:

$$F(a + 2\Delta x) \approx F(a + \Delta x) + _____$$

$$\approx F(a) + _____\Delta x + _____\Delta x$$

- Use $F'(a + 2\Delta x) = f(a + 2\Delta x)$ to approximate $F(a + 3\Delta x)$:

$$F(a + 3\Delta x) \approx F(a + 2\Delta x) + _____$$

$$\approx F(a) + _____\Delta x + _____\Delta x + _____\Delta x$$

$$= F(a) + \overline{\sum_{i=____}} _____\Delta x$$

- Find a sum that approximates $F(a + n\Delta x)$:

$$F(a + n\Delta x) \approx F(a) + \overline{\sum_{i=____}} _____\Delta x$$

If $\Delta x = \frac{b-a}{n}$, then $a + n\Delta x =$ ___, and the sum in the last bullet point above is exactly _____ for $\int_a^b f(x)\,dx$:

$$F(___) \approx F(a) + _____.$$

As the step size Δx decreases, we expect that the linear approximation gets better and better. Therefore, it makes sense that if we apply $\lim_{n\to\infty}$ so that $\Delta x \to 0$, the $\approx$ will become an $=$. We thus get the Fundamental Theorem of Calculus.

The Fundamental Theorem of Calculus:
If $f(x)$ is a continuous function on the interval $[a, b]$ and $f(t) = F'(t)$, then

$$\int_a^b f(t)\,dt = F(b) - F(a)$$

Examples

1. Use the FTC to compute the first two definite integrals from page 3. Make sure your □Regrade answers match up with what you got there. (We'll do the last one in a bit.)

2. Find the area under the curve $f(x) = x^2$ between $x = 1$ and $x = 4$ exactly (Finally!). □Regrade Compare your answers to the over and underestimates from question 5.

3. Compute the integral: $\int_1^2 e^{-x}\,dx$. Compare your answer to the estimates in Part II □Regrade of the Riemann Sums lab.

(Hint: what is an antiderivative of e^{-x}? You want a function whose derivative is e^{-x}.)

4. ☐Regrade (a) Consider the function $f(x) = \frac{1}{x^2}$. Draw a graph of this between $x = -1$ and $x = 1$. Then shade in the area corresponding to the integral $\int_{-1}^{1} \frac{1}{x^2}\, dx$.

(b) Why doesn't the FTC help with computing this integral? (Hint: look at the conditions of the theorem.)

(c) Try computing it using the FTC anyway. How do you know the answer is incorrect?

5. ☐Regrade Can you compute the integral: $\int_{1}^{2} e^{-x^2}\, dx$? Why or why not? (Hint: You'll need an antiderivative. If you think you have found one, make sure you differentiate to check it...)

6. ☐Regrade (a) What is $\int_{-1}^{1} \sqrt{1-x^2}\, dx$? (Copy your answer from when we solved this earlier in the sheet.)

(b) Show that $F(x) = \frac{1}{2}x\sqrt{1-x^2} + \frac{1}{2}\arcsin x$ is an antiderivative of $f(x) = \sqrt{1-x^2}$. (Hint: show that $F'(x) = f(x)$. The algebra here is hard. If you get stuck, this isn't the most important idea here – assume that $F(x)$ is an antiderivative and move on to the next question.)

(c) Use the previous part to compute $\int_{-1}^{1} \sqrt{1-x^2}\, dx$.

WARNING: While the FTC is a very powerful theorem that allows us to compute the area under many curves, it isn't helpful in the following situations:

- If $f(x)$ is not _____, the FTC does not apply (see question 11 above).
- If we cannot find an _____ for $f(x)$, we can't apply the FTC in the first place (see question 12 above).
- Sometimes it's just easier to compute an area geometrically (see question 13 above or the questions on page 2).